Abstract

The present invention discloses a system and method for automatic multi-level revenue allocation of cognitive assets based on smart contracts. The system comprises a **dependency graph tree construction unit**, a multi-path cascaded bookkeeping and clearing unit, a **dynamic weight allocation module**, a **lifecycle state transition contract module**, and a **secure pull-type withdrawal unit**. When a sub-protocol is registered, the system parses parent fingerprints, recursively traces to generate a derived tree topology graph, and implements interception with a hard maximum inheritance depth limit of **18 generations**. Upon commercial settlement, the system combines optimistic locks and timestamp verification to write transaction funds into state mapping variables in a cascaded manner according to an asymmetric gradient ratio of **75%/10%/15%**. If a sub-protocol inherits multiple parent protocols, the system calculates dynamic allocation weights based on fingerprint similarity. When a node has existed for **20 years** or is a **root-level meta-protocol**, split revenue is automatically redirected by a state machine to a hard-coded, tamper-proof **ecological directional backfeed pool**. External payees actively initiate withdrawal requests with public-key encryption and anti-replay **Nonce** verification, and complete two-stage pull-type extraction to defend against reentrancy attacks. While protecting multi-level property rights, the present invention reduces virtual machine Gas overhead and concurrent conflicts.

Figure 1 is designated as the accompanying drawing for the abstract.